Specular reflection model for nonlocality

This set of notes is on the paper Polarizability of a small sphere including nonlocal effects (https://journals.aps.org/prb/pdf/10.1103/PhysRevB.24.554) by **Dasgupta and Fuchs**. The results derived in this paper are used in the following paper by **de Abajo**: Spatial Nonlocality in the Optical Response of Metal Nanoparticles (https://pubs.acs.org/doi/pdf/10.1021/jp204261u). The crux of the method by **Dasgupta and Fuchs** (at least, as I see it) is to map the problem of calculating surface effects to a sister problem: imagining the entirety of space to be occupied by the nonlocal material with permitivity $\varepsilon(k,\omega)$ with an external surface charge density at the boundary of the sphere. The equations derived for the potentials inside the sphere (i.e. for radial coordinates smaller than the sphere radius, $r<a$) will be assumed to be correct– after which, matching conditions will be used to derive the polarizability.

In the following sections, we will derive the key formulas of the paper by **Dasgupta and Fuchs**. Note that any notation not defined in the below is implicitly understood to be the notation of the paper by **Dasgupta and Fuchs**.

Section 1: Derivation of Equation 11

We write Laplace's equation for the potential associated with the displacement field (as defined in the paper):

$$\nabla^2\left[\frac{1}{(2\pi)^3}\int V_D(\mathbf{k})e^{i\mathbf{k}\cdot\mathbf{r}}\mathrm{d}^3\mathbf{k}\right] = -\sigma(\theta)\delta(r-a),$$

where $\sigma(\theta)$ is the effective external charge density per unit area. Note that we take $\sigma(\theta)$ to be azimuthally symmetric since we are implicitly dealing with potentials of the form introduced in **Equations 9a/9b** of the paper. Since the external charge density must be related to the change in the displacement field across the boundary, we must have:

$$\sigma(\theta) = -\left[\partial_r V_D(\mathbf{r})\right]\Big|_{r=a^+} + \left[\partial_r V_D(\mathbf{r})\right]\Big|_{r=a^-} = \cos(\theta_{r,E})\delta,$$

where δ is defined by **Equation 12** of the paper. By intoducing the form of the charge density into the Laplace equation, we obtain the following:

$$\frac{1}{(2\pi)^3}\int V_D(\mathbf{k})|\mathbf{k}|^2 e^{i\mathbf{k}\cdot\mathbf{r}}\mathrm{d}^3\mathbf{k} = \delta\cos(\theta_{r,E})\delta(r-a) \rightarrow \text{ integrate both sides with } e^{-i\mathbf{k}'\cdot\mathbf{r}} \rightarrow$$
$$V_D(\mathbf{k})|\mathbf{k}|^2 = \delta\int e^{-i\mathbf{k}\cdot\mathbf{r}}\cos(\theta_{r,E})\delta(r-a)\mathrm{d}^3\mathbf{r} = a^2\delta\int_{r=a} e^{-i\mathbf{k}\cdot\mathbf{r}}\cos(\theta_{r,E})\mathrm{d}\Omega,$$

which is **Equation 11**, as desired.

Section 2: Derivation of Equations 14/15

In order to derive **Equation 14** of the paper, we write $\cos(\theta_{r,E}) = \sqrt{\frac{4\pi}{3}}Y_{l,0}(\theta_{r,E},\phi_{r,E})$, which gives us:

$$a^2\delta\int_{r=a} e^{-i\mathbf{k}\cdot\mathbf{r}}\cos(\theta_{r,E})\mathrm{d}\Omega = a^2\delta\sqrt{\frac{4\pi}{3}}(-4\pi i)j_1(ka)Y_{1,0}(\theta_{k,E},\phi_{k,E}) = -4\pi i a^2\delta j_1(ka)\cos(\theta_{k,E}),$$

where $\Omega = \sin(\theta_{r,E})\mathrm{d}\theta_{r,E}\mathrm{d}\phi_{r,E}$ is the solid angle in real space. Note that we will use the same notation to denote the solid angle in momentum space, but it will be clear from context whether Ω refers to a real space or a momentum space quantity. With this digression aside, we obtain the form of the potential in momentum space as follows:

$$V_D(\mathbf{k}) = -\frac{4\pi i a^2\delta}{k^2}j_1(ka)\cos(\theta_{k,E}),$$

which is **Equation 14** (***Note that I believe the paper is off by a minus sign, but this error does not affect the result for the polarizability***). Note that, furthermore, **Equation 15** follows readily from the definition of the dielectric function.

Section 3: Derivation of Equations 16/17

Equation 16 and **Equation 17** give the forms of the potentials in real space, for, crucially, $r < a$. To go back into real space, we must Fourier transform the momentum space potentials derived in the previous section:

$$V_D(\mathbf{r}) = -\frac{1}{(2\pi)^3}\int\frac{4\pi i a^2\delta}{k^2}j_1(ka)\cos(\theta_{k,E})e^{i\mathbf{k}\cdot\mathbf{r}}k^2\mathrm{d}\Omega\mathrm{d}k =$$
$$\frac{(4\pi)^2}{8\pi^3}a^2\delta\cos(\theta_{r,E})\int_0^\infty j_1(ka)j_1(kr)\mathrm{d}k,$$

where we used the plane wave to spherical wave transformation given by **Equation 13** (which we also used in the previous section). From https://functions.wolfram.com/Bessel-TypeFunctions/SphericalBesselJ/21/02/02/0001/ (https://functions.wolfram.com/Bessel-

TypeFunctions/SphericalBesselJ/21/02/02/0001/), we see that the Bessel function integral may be cast as follows:

$$\int_0^\infty j_1(ka)j_1(kr)\mathrm{d}k = \frac{\pi a^{-2}r}{6}(\text{for } r < a) \to V_D(\mathbf{r}) = \frac{\delta\cos(\theta_{r,E})r}{3},$$

which is **Equation 16**.

Similarly, we have:

$$V(\mathbf{r}) = \frac{2a^2\delta}{\pi}\cos(\theta_{r,E})\int_0^\infty \frac{j_1(ka)j_1(kr)}{\varepsilon(k,\omega)}\mathrm{d}k,$$

which is **Equation 17**.

Section 4: Derivation of Equation 19

The results of the previous sections give us expressions for the potentials/fields in the medium. In order to find the polarizability, we must match these internal fields appropriately with the fields outside the sphere (written in terms of the applied field and the dipole moment). The continuity of the normal component of the displacement field gives us:

$$-(\delta/3) = E_0 + 2p/a^3$$

The continuity of the potential (equivalently the tangential component of the electric field) gives us:

$$-E_0 + p/a^3 = a\delta F(a) = -a\left[3E_0 + 6p/a^3\right]F(a)$$

We now write this relation in terms of the polarizability, $\alpha \equiv p/E_0$:

$$-1 + \alpha/a^3 = -a\left[3 + 6\alpha/a^3\right]F(a) \to \frac{\alpha}{a^3} = \frac{1-3aF(a)}{1+6aF(a)},$$

which is **Equation 19** of the paper.